

Computing Machinery and Intelligence

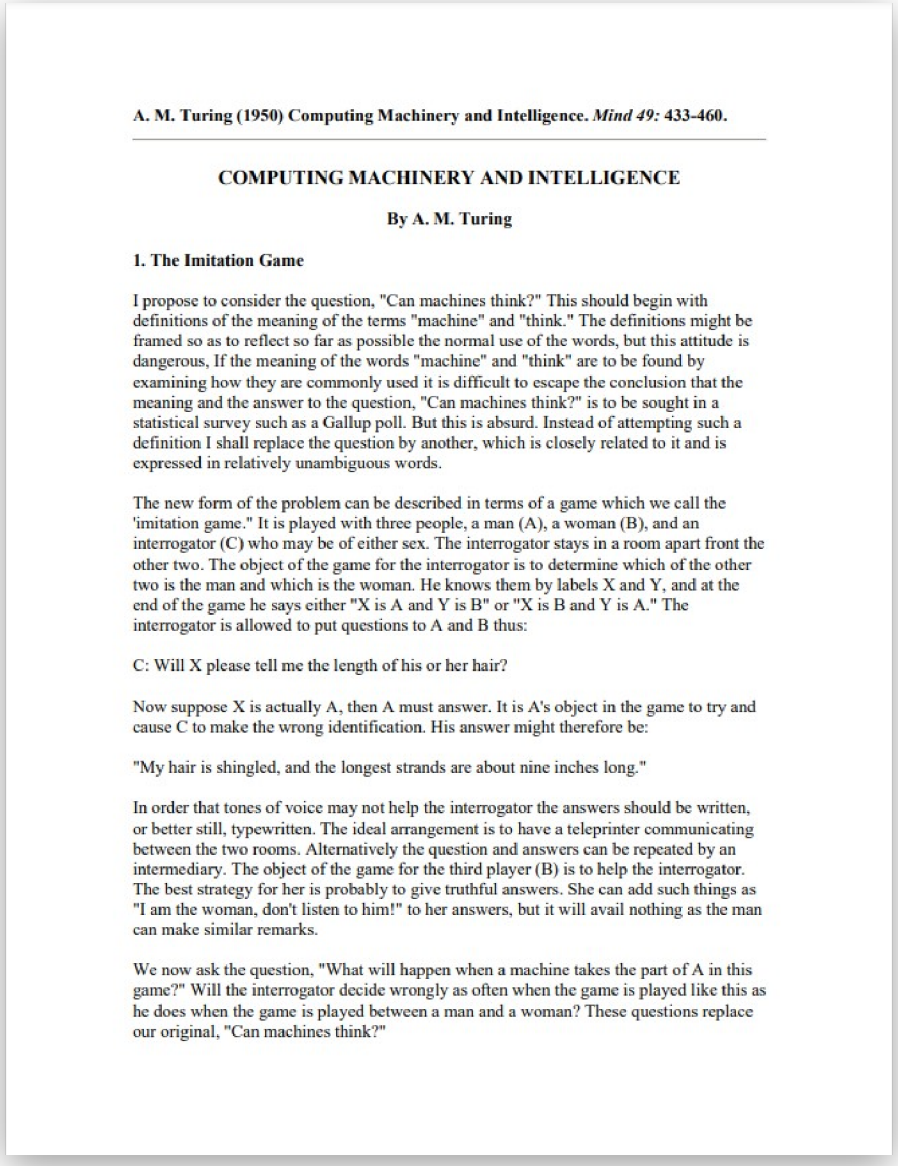
Can machines think?

Alan Mathison Turing
Mind, , 1950
Oxford University - Oxfordshire, UK.

Computing Machinery and Intelligence

Can machines think?

Alan Mathison Turing
Mind, , 1950
Oxford University - Oxfordshire, UK.



A. M. Turing (1950) Computing Machinery and Intelligence. *Mind 49:* 433-460.

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous, If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game." It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart front the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either "X is A and Y is B" or "X is B and Y is A." The interrogator is allowed to put questions to A and B thus:

C: Will X please tell me the length of his or her hair?

Now suppose X is actually A, then A must answer. It is A's object in the game to try and cause C to make the wrong identification. His answer might therefore be:

"My hair is shingled, and the longest strands are about nine inches long."

In order that tones of voice may not help the interrogator the answers should be written, or better still, typewritten. The ideal arrangement is to have a teleprinter communicating between the two rooms. Alternatively the question and answers can be repeated by an intermediary. The object of the game for the third player (B) is to help the interrogator. The best strategy for her is probably to give truthful answers. She can add such things as "I am the woman, don't listen to him!" to her answers, but it will avail nothing as the man can make similar remarks.

We now ask the question, "What will happen when a machine takes the part of A in this game?" Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, "Can machines think?"

A. M. Turing (1950) **Computing Machinery and Intelligence**.
Mind 49: 433-460.

Lists

1. London
2. Manchester
3. Birmingham
4. Glasgow
5. Liverpool

1. Newcastle
2. Sheffield
3. Nottingham
4. Southampton
5. Edinburgh

Is Algebraic Graph Knowledge Possible?

Research has been conducted in order to evaluate the possibility of reaching meaningful knowledge from Algebraic Graph transformations.

- Model Cheking and theorem prooving are viable paths.

When the neet to make strong assertions becomes inevitable:

- This is the first way: outstanding assertion !
- Even greater impact comes from: **hilight text** !

> ***Note** : This is a very long footnote line intended to test the layout of two.

H1

H2

H3

H4

H5

H6

- This is a fragment o normal text written here in order to exemplify the use of several featrues in CSS.
- This is a fragment o normal text written here in order to exemplify the use of several featrues in CSS.
- This is one **feature**
- This is another subjetc.

Lists

1. One
2. Two
3. Three
    i. abc
    ii. def
4. End of list

```haskell
primes = filterPrime [2..]
  where filterPrime (p:xs) =
          p : filterPrime [x | x <- xs, x `mod` p /= 0]

seqLength :: Num b ⇒ Sequence a → b
seqAppend :: Sequence a → Sequence a → Sequence a

seqLength Nil = 0
seqLength (Cons _ xs) = 1 + seqLength xs

seqAppend Nil ys = ys
seqAppend (Cons x xs) ys = Cons x (seqAppend xs ys)
```

Code: Haskell code fragment.

Tables

Column A	Column B	Column C	Column D
A1	B1	C1	D1
A2	B2	C2	D2
A3	B3	C3	D3

Table: Exemple of use of tables.

LaTeX Equations

$$\frac{1}{c^2}\frac{\partial^2 \psi}{\partial t^2} = \nabla^2 \circ \psi$$

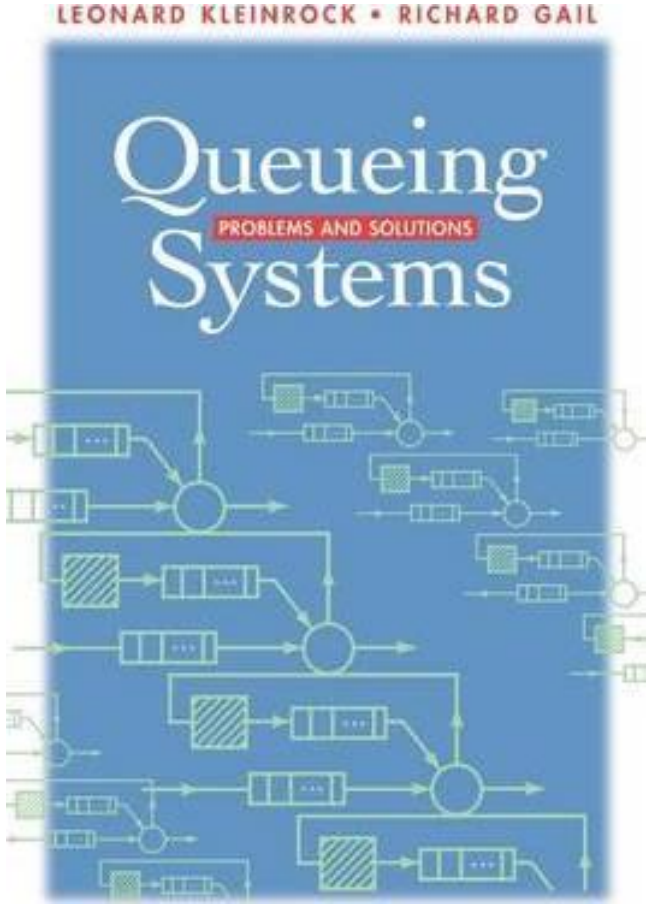
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

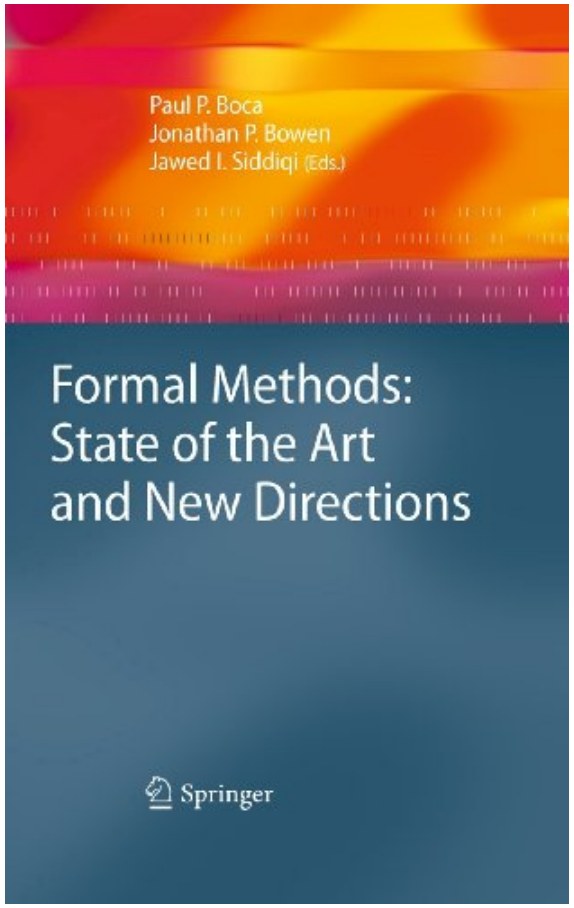
$$\nabla^2 \mathbf{E} = \mu\epsilon\frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$c = \sqrt{\frac{1}{\mu\epsilon}}$$

Formulae: Exemples of LaTeX formulas use.

Images in Two Columns





Images in Two Columns

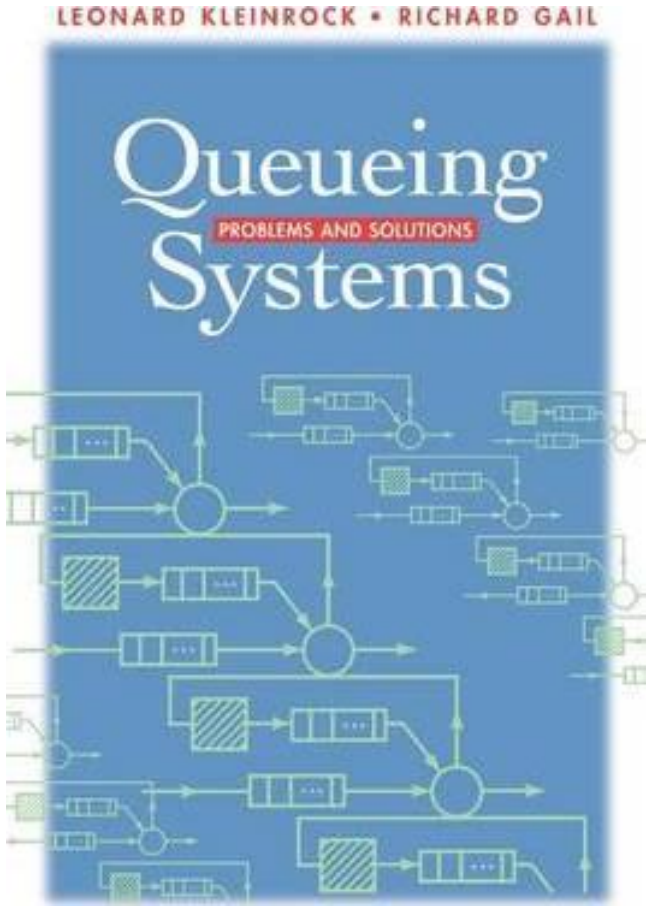


Figure: Kleinrock, Gail (1979).

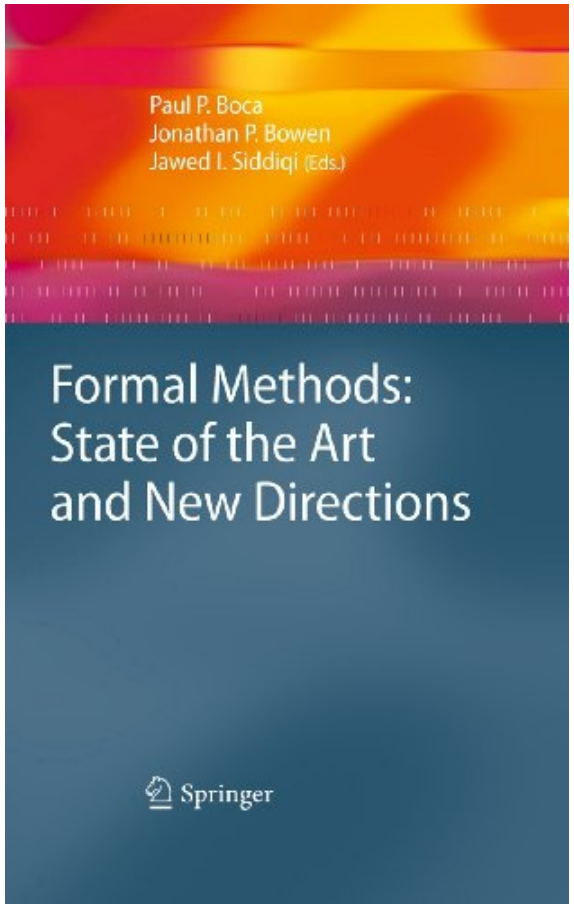


Figure: Springer Verlag (1979).

Image and text

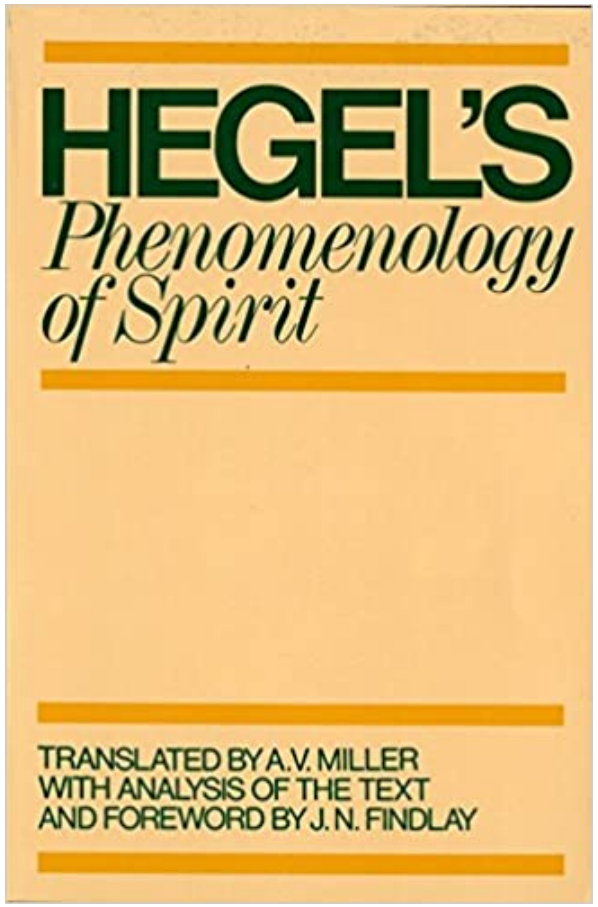


Figure: Oxford edition (1979).

Hegel's Phenomenology

The book was originally entitled "Phänomenologie des Geistes" by its author, G.W.F. Hegel.

- Published in 1807, marked a significant development in German idealism after Kant.
- In this book Hegel develops his concepts of dialectic.

Price at Amazon: $ 17.83

"There is an increasing demand of current information systems to incorporate the use of a higher degree of formalism in the development process. Formal Methods consist of a set of tools and techniques based on mathematical model and formal logic that are used to specify and verify requirements and designs for hardware and software systems."

"There is an increasing demand of current information systems to incorporate the use of a higher degree of formalism in the development process. Formal Methods consist of a set of tools and techniques based on mathematical model and formal logic that are used to specify and verify requirements and designs for hardware and software systems."

- Mona Batra -

Transition Slide 1

Aditional Text

Transition Slide 2

Aditional Text



References

1. PLATO. Plato Republic . Tradução: C. D. C. Reeve. Indianapolis, IN, USA: Hackett Publishing Company, 2004.
2. PLATO. Plato Republic . Tradução: C. D. C. Reeve. Indianapolis, IN, USA: Hackett Publishing Company, 2004.
3. ARISTOTELES. Nikomachische Ethik . Berlin: Akademie Verlag, 2010. (Klassiker Auslegen).v. 2
4. KANT, Immanuel. Kritik der Praktischen Vernunft . Berlin: Akademie Verlag, 2002. (Klassiker Auslegen).v. 26
5. HEGEL, Georg Friederich Wilhelm. Hegel´s Phenomenology of Spirit . Tradução: A. V. Miller. New York: Oxford University Press, 2004.

References

1. PLATO. **Plato Republic**. Tradução: C. D. C. Reeve. Indianapolis, IN, USA: Hackett Publishing Company, 2004.
2. ARISTOTELES. **Nikomachische Ethik**. Berlin: Akademie Verlag, 2010. (Klassiker Auslegen).v. 2
3. KANT, Immanuel. **Kritik der Praktischen Vernunft**. Berlin: Akademie Verlag, 2002. (Klassiker Auslegen).v. 26
4. HEGEL, Georg Friederich Wilhelm. **Hegel´s Phenomenology of Spirit**. Tradução: A. V. Miller. New York: Oxford University Press, 2004.
5. HUSSERL, Edmund. **The Crisis of European Sciences and Transcendental Phenomenology**. Evanston, USA: Northwestern University Press, 1970.
6. CASSIRER, Ernst. **The Myth of the State**. New Haven, USA: Yale University Press, 1946.
7. HEIDEGGER, Martin. **Sein und Zeit**. 11. ed. Tübingen: Max Niemeyer Verlag, 1967.
8. GADAMER, Hans-Georg. **Wahrheit und Methode**. Berlin: Akademie Verlag, 2007. v. 30.